

them initially, longer time horizons and higher initial energies are usually necessary. Once an initial condition that gives rise to a nonlinear optimal is found, this resulting nonlinear optimal can in turn also be used as an initial conditions for calculations with different parameters. Most of the parameter combinations – including all that we focus the following discussion on – were also rerun from different initial conditions to ensure that the same optimal was reached, although, since this is a nonlinear optimisation problem, there is no rigorous way to ensure that a global optimum has indeed been found. Nevertheless, the conclusions made in the following are quite robust to changes in the computational setup, including the domain size, as is further discussed in Appendix A.2.

Since the code written for this study supports automatic differentiation, it would also be possible to obtain the necessary gradient information without solving the adjoint system. However, it has been found that using the classical adjoint-based approach performs better than automatic differentiation, both in terms of memory and runtime. In particular, the ability to more easily control the trade-off between memory usage and runtime through checkpointing makes the adjoint approach preferable since we are currently already close to the limits of available GPU-memory on an NVIDIA A100 card with 40 GB memory. Moreover, the gradients obtained from the adjoint method are fairly smooth, whereas in gradients obtained from automatic differentiation, the discretisation error manifests itself in the form of noise, so that some processing of these gradients would likely be necessary. Automatic differentiation does have the advantage of requiring less development time and thus being more readily applied to new problems, so that this feature will likely be valuable in future studies.

4. Results

As discussed in Section 2, the main parameters are the time horizon T and the initial disturbance energy e_0 . Unlike in the transition problem, it is not useful to look at very large time horizons, as the timescale on which disturbances can grow is constrained by large-scale disruptive turbulent events. This argument has been used by Butler & Farrell (1993) to determine the timescale for linear optimal growth, and the general idea also applies to the nonlinear case. From previous investigations (Lozano-Durán *et al.* 2021; Markeviciute & Kerswell 2024), we can estimate that the primary and the secondary linear process both operate on a timescale of around $t_p = t_s = 0.35h/u_\tau$, and that bursting events are roughly $t_{\text{burst}} = 4h/u_\tau$ apart. This motivates an initial focus on the time horizon $T = 0.7h/u_\tau = 2t_p$, as this would be enough to see both the primary and the secondary linear process. This is addressed in Section 4.1. To get a deeper understanding of the role of this parameter, we additionally study times $t_p, 1.5t_p, 3t_p, 4t_p, 6t_p$ and $8t_p$ in Section 4.2.

For any T , as $e_0 \rightarrow 0$, the linear optimisation problem is recovered. Consequently, for small e_0 , a quasilinear optimal, which closely resembles the linear optimal, is obtained. As e_0 increases, there comes a point where a structurally completely different nonlinear optimal arises. For large e_0 , the final state of the disturbances may be well in the turbulent regime, which, in the current formulation, usually leads the optimisation algorithm to fail, as the final state dependency on the initial condition is too sensitive and chaotic for the optimisation problem to remain well-conditioned. The precise determination of this boundary is difficult and not very important to the present study, but values up to $e_0/E_0 = 10^{-4}$ were found to converge without issues even for the longest time horizons considered here, where E_0 denotes the energy of the base flow. Naturally, we are most interested in nonlinear optimals whose initial energy is below this value.